

# R&D Model

Note: This content will help you once you have gone through the book. If you did not read this model from the book before, you may face some trouble. In that case, check that particular topic from the book or simply ask someone to explain the issue who have the habit of reading books. Also do not use any unfair means in the exam using this content (or any content).

# R&D Model

## 3.1 Framework and Assumptions

### Overview

- To model the accumulation of knowledge, we need to introduce a separate sector of the economy where **new ideas are developed**.
- Which is *research and development* (or *R&D*)

Two major simplifications:

- **First**, both the R&D and goods production functions are assumed to be **generalized Cobb–Douglas functions**;
  - that is, they are power functions, but the sum of the exponents on the inputs is not necessarily restricted to 1.
- **Second**, R&D model takes the fraction of **output saved** and the fractions of **the labor force and the capital stock** used in the R&D sector as **exogenous** and **constant**.

### Specifics

- The model, involves four variables:
  - labor ( $L$ ), capital ( $K$ ), technology ( $A$ ), and output ( $Y$ ).
- The model is set in continuous time.
- There are two sectors,
  - **A goods-producing sector** where **output** is produced and
  - **An R&D sector** where **additions to the stock of knowledge** are made.
- Fraction  $a_L$  of the labor force is used in the **R&D sector**
- fraction  $1-a_L$  in the **goods-producing sector**.
- Similarly, fraction  $a_K$  of the capital stock is used in **R&D** and
- the rest  $1-a_K$  in **goods production**.
- Both  $a_L$  and  $a_K$  are **exogenous** and **constant**.
- Because the use of an idea or a piece of knowledge in one place does not prevent it from being used elsewhere, **both sectors** use the **full stock of knowledge**,  $A$ .
- The quantity of output produced at time  $t$  is thus

$$Y(t) = [(1 - a_K)K(t)]^\alpha [A(t)(1 - a_L)L(t)]^{1-\alpha} \quad 0 < \alpha < 1 \quad (3.1)$$

- Equation (3.1) implies **constant returns** to capital and labor.

- Given our assumption of generalized Cobb–Douglas production, we can write

$$\dot{A}(t) = B [a_K K(t)]^\beta [a_L L(t)]^\gamma A(t)^\theta \quad B > 0, \beta \geq 0, \gamma \geq 0, \quad (3.2)$$

where  $B$  is a shift parameter.

- The production function for knowledge is **not assumed to have constant returns** to scale to capital and labor.

(You can ignore this if you want)

The standard argument that:

- there must be at least constant returns is a replication one:
  - if the inputs double, the new inputs can do exactly what the old ones were doing, thereby doubling the amount produced.
- But in the case of knowledge production, exactly replicating what the existing inputs were doing would cause the same set of discoveries to be made twice, thereby leaving  $\dot{A}$  unchanged.
- **Thus it is possible that there are diminishing returns in R&D.**

At the same time,

- interactions among researchers, fixed setup costs, and so on may be important enough in R&D that
  - Doubling capital and labor more than doubles output.
- **We therefore also allow for the possibility of increasing returns.**

- The parameter  $\theta$  reflects the effect of the existing stock of knowledge on the success of R&D.
  - This effect can operate in either direction.
- On the one hand, **past discoveries may provide ideas and tools** that make future discoveries easier.
  - In this case,  **$\theta$  is positive.**
- On the other hand, **the easiest discoveries may be made first.**
  - In this case, it is harder to make new discoveries when **the stock of knowledge is greater**, and
  - so  **$\theta$  is negative.**

Because of these conflicting effects, no restriction is placed on  $\theta$  in (3.2).

- The **saving** rate is **exogenous** and **constant**. (Similar to Solow)
- In addition, **depreciation** is set to **zero** for simplicity. (Similar to RCK)

Thus,

$$\dot{K}(t) = sY(t) \quad (3.3)$$

- Likewise, we continue to treat population growth as exogenous and constant.
  - For simplicity, we do not consider the possibility that it is negative.

This implies

$$\dot{L}(t) = nL(t) \quad n \geq 0 \quad (3.4)$$

- Finally, as in our earlier models, the initial levels of  $A$ ,  $K$ , and  $L$  are given and strictly positive.
- Because the model has two state variables whose behavior is endogenous,  $K$  and  $A$ , it is more complicated to analyze than the Solow model.
- We therefore begin by considering **the model without capital**; that is, we set  $\alpha$  and  $\beta$  to zero

## 3.2 The Model without Capital

### The Dynamics of Knowledge Accumulation

When there is no capital in the model, the production function for output (equation [3.1]) becomes

$$Y(t) = A(t) (1 - a_L)L(t) \quad (3.5)$$

Similarly, the production function for new knowledge (equation [3.2]) is now

$$\dot{A}(t) = B[a_L L(t)]^\gamma A(t)^\theta \quad (3.6)$$

Population growth continues to be described by equation (3.4).

- Equation (3.5) implies that output per worker is proportional to  $A$ .
  - Thus the growth rate of output per worker equals the growth rate of  $A$ .
  - We therefore focus on the **dynamics of  $A$** , which are given by (3.6).
  - This equation implies that the growth rate of  $A$ , denoted  $g_A$ , is

$$\begin{aligned} g_A(t) &\equiv \frac{\dot{A}(t)}{A(t)} \\ &= B a_L L(t)^\gamma A(t)^{\theta-1} \end{aligned} \quad (3.7)$$

- Taking logs of both sides of (3.7) and differentiating the two sides with respect to time gives us an expression for the **growth rate of  $g_A$** ,

$$\frac{\dot{g}_A(t)}{g_A(t)} = \gamma n + (\theta - 1) g_A(t) \quad (3.8)$$

Multiplying both sides of this expression by  $g_A(t)$  yields

$$\dot{g}_A(t) = \gamma n g_A(t) + (\theta - 1)[g_A(t)]^2 \quad (3.9)$$

• To describe further how the growth rate of  $A$  behaves (and thus to characterize the behavior of output per worker), we must distinguish among the cases

- $\theta < 1$
- $\theta > 1$  and
- $\theta = 1$

### Case 1: $\theta < 1$

Figure 3.1 shows the phase diagram for  $g_A$  when  $\theta$  is less than 1. That is, it plots  $\dot{g}_A$  as a function of  $A$  for this case.

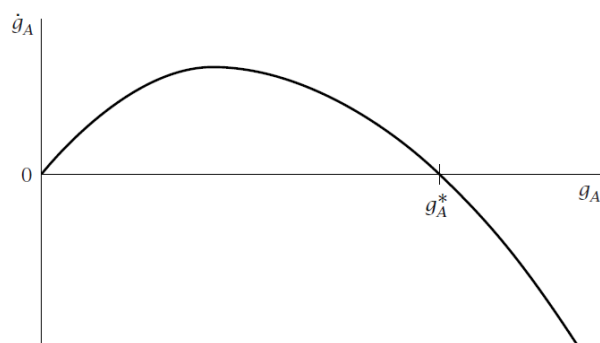


FIGURE 3.1 The dynamics of the growth rate of knowledge when  $\theta < 1$

- Equation (3.9) implies that
  - $\dot{g}_A$  is **positive** for **small** positive values of  $g_A$
  - **Negative** for **large** values.
- We will use  $g_A^*$  to denote the unique positive value of  $g_A$  that implies that  $\dot{g}_A$  is zero.
- From (3.9),  $g_A^*$  is defined by  $\gamma n + (\theta - 1)g_A^* = 0$

Solving this for  $g_A^*$  yields

$$g_A^* = \frac{\gamma}{1-\theta} n \quad (3.10)$$

- This analysis implies that regardless of the economy's initial conditions,  $g_A$  converges to  $g_A^*$
- If the parameter values and the initial values of  $L$  and  $A$  imply  $g_A(0) < g^*$ 
  - $\dot{g}_A$  is **positive**
  - that is,  $g_A$  is **rising**
  - It continues to rise **until it reaches  $g_A^*$**

- Similarly, if  $g_A(0) > g_A^*$ 
  - then  $g_A$  **falls** until it reaches  $g_A^*$
- Once  $g_A$  reaches  $g_A^*$ 
  - both  $A$  and  $\frac{Y}{L}$  grow steadily at rate  $g_A^*$
- Thus the economy is on a **BGP**.

This model is our first example of a model of *endogenous growth*.

- The model implies that the long-run growth rate of output per worker,  $g_A^*$  is an **increasing function** of the rate of population growth,  $n$ .
- Equation (3.10) also implies that the **fraction of the labor** force engaged in R&D **does not affect** long-run growth.

### To see why it does not:

- Suppose there is a **permanent increase in  $a_L$**  starting from a situation
  - where  $A$  is growing at rate  $g_A^*$
- $a_L$  does not enter expression (3.9) for  $\dot{g}_A$
- Thus the rise in  $a_L$  **does not affect** the curve showing  $\dot{g}_A$  as a function of  $g_A$ .
- But  $a_L$  does enter expression (3.7) for  $g_A$
- The increase in  $a_L$  therefore
  - causes an **immediate increase** in  $g_A$
  - but **no change in  $\dot{g}_A$**  as a function of  $g_A$ .

This is shown by the dotted arrow in Figure 3.2.

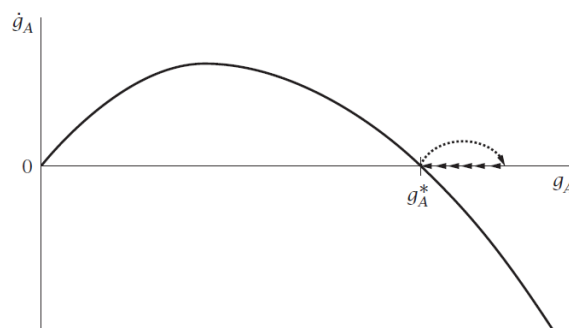


FIGURE 3.2 The effects of an increase in  $a_L$  when  $\theta < 1$

- As the phase diagram shows, the increase in the growth rate of knowledge is not sustained.
- When  $g_A$  is above  $g_A^*$ 
  - $\dot{g}_A$  is **negative**
  - $g_A$  therefore **returns gradually to  $g_A^*$**  and then remains there.

- This is shown by the **solid arrows** in the figure.
- Intuitively, the fact that  $\theta$  is less than 1 means that
  - The contribution of additional knowledge to the production of new knowledge is **not strong enough to be self-sustaining**.
- This analysis implies that
  - the **increase in  $a_L$**  results in a **rise in  $g_A$**  followed by a **gradual return** to its **initial level**.
- That is, it has a **level effect** but **not a growth effect** on the path of  $A$ .

This information is summarized in Figure 3.3

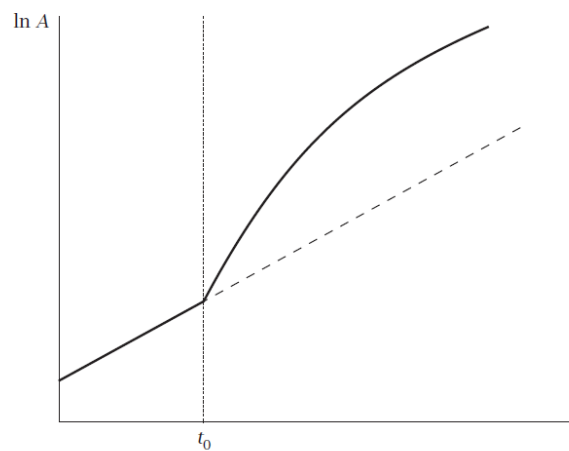


FIGURE 3.3 The impact of an increase in  $a_L$  on the path of  $A$  when  $\theta < 1$

### Case 2: $\theta > 1$

Equation (3.9):  $\dot{g}_A = \gamma n g_A + (\theta - 1) g_A^2$

- When  **$\theta$  exceeds 1**, equation (3.9) implies that
  - $\dot{g}_A$  is **positive** for all possible values of  $g_A$
  - Further, it implies that  $\dot{g}_A$  is **increasing** in  $g_A$
  - The phase diagram is shown in Figure 3.4.

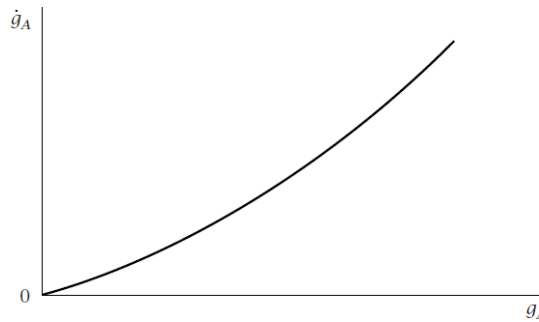


FIGURE 3.4 The dynamics of the growth rate of knowledge when  $\theta > 1$

- As the phase diagram shows, the economy exhibits **ever-increasing growth** rather than convergence to a BGP.
- Thus once the accumulation of knowledge begins the economy embarks on a path of **ever-increasing growth**.

The impact of an increase in the fraction of the labor force engaged in R&D is now dramatic.

- From Equation (3.7) an **increase in  $a_L$** 
  - causes an immediate **increase** in  $g_A$ , as before.
  - But  $\dot{g}_A$  is an increasing function of  $g_A$
  - thus  $\dot{g}_A$  **rises** as well.
- And the **more rapidly  $g_A$  rises**, the **more rapidly its growth rate rises**.
- Thus the **increase in  $a_L$**  causes the growth rate of  $A$  to exceed what it would have been otherwise by **an ever-increasing amount**.

### Case 3: $\theta = 1$

- When  $\theta$  is exactly equal to 1
  - existing knowledge is **just productive enough** in generating new knowledge
  - that the **production of new knowledge** is **proportional** to **the stock**.

In this case, expressions (3.7) and (3.9) for  $g_A$  and  $\dot{g}_A$  simplify to

$$g_A(t) = B a_L^\gamma L(t)^\gamma \quad (3.11)$$

$$\dot{g}_A(t) = \gamma n g_A(t) \quad (3.12)$$

- If **population growth is positive**
  - $g_A$  is growing over time
  - in this case the dynamics of the model are similar to those when  $\theta > 1$ .
- If population growth is zero
  - $g_A$  is **constant** regardless of the initial situation.
  - Thus there is **no adjustment toward a BGP**:
  - No matter where it begins, the economy immediately exhibits **steady growth**.



- As equations (3.5) and (3.11) show, the growth rates of knowledge, output, and output per worker are all equal to  $B\alpha_L^\gamma L^\gamma$  in this case.
- Thus **changes in  $\alpha_L$  affect the long-run growth rate** of the economy.

### How the saving rate affects long-run growth

- Since the output good in this economy has **no use** other than in **consumption**
    - it is natural to think of it as being **entirely consumed**.
  - Thus  $1 - \alpha_L$  is the fraction of society's resources **devoted to producing goods** for current consumption,
  - and  $\alpha_L$  is the fraction devoted to producing a good (namely, **knowledge**) that is useful for producing output in the future.
  - Thus one can think of  $\alpha_L$  as **a measure of the saving rate** in this economy.
- ❖ With this interpretation, **the case of  $\theta = 1$  and  $n = 0$**  provides a simple example of a **model where the saving rate affects long-run growth**.
  - ❖ Models of this form are known as **linear growth models**
  - ❖ They are also known as  **$Y = AK$  models**

### **The Importance of Returns to Scale to Produced Factors**



- The **reason** that the three cases have such different implications is that
  - whether  $\theta$  is **less than**, **greater than**, or **equal to 1** determines whether there are **decreasing**, **increasing**, or **constant returns to scale** to **produced factors of production**.
- The growth of labor is exogenous, and we have **eliminated capital** from the model
  - thus **knowledge is the only produced factor**.
- There are **constant returns** to knowledge **in goods production**.
- Thus whether there are **on the whole increasing**, **decreasing**, or **constant returns** to knowledge in this economy
  - is **determined by the returns to scale to knowledge** in knowledge production
  - that is, by  $\theta$ .

### **The Importance of Population Growth**

#### When $\theta < 1$

- The model has the surprising implication that
  - **Positive population growth** is necessary for **long-run growth** in **income per person**
  - the economy's **long-run growth rate** is **increasing** in **population growth**.

The other cases have similar implications.

### When $\theta = 1$ and $n = 0$

- long-run growth is an **increasing** function of the **level** of population.

### When $\theta > 1$ (or $\theta = 1$ and $n > 0$ )

- one can show that an **increase in population** growth causes
  - **income per person** to be **higher** than it otherwise would have been by an **ever-increasing amount**.

#### To understand these results,

- Consider equation (3.7) for knowledge accumulation:  $g_A(t) = B a_L L(t)^\gamma A(t)^{\theta-1}$
- Built into this expression is the completely natural idea that
  - when there are more people to make discoveries, more discoveries are made.
- And when more discoveries are made
  - the stock of knowledge grows faster, and
  - so (all else equal) output per person grows faster.

A **natural interpretation of the model** is that

- $A$  represents knowledge that can be used anywhere in the world.
- With this interpretation,
  - the model does not imply that countries with larger populations, or countries with greater population growth, enjoy greater income growth
  - it only implies that **higher worldwide population growth raises worldwide income growth**.
  - Because higher population growth is beneficial to the growth of worldwide knowledge.

### **3.3 The General Case**

We now want to reintroduce capital into the model

#### **The Dynamics of Knowledge and Capital**

- When the model includes capital, there are **two endogenous** state variables,  $A$  and  $K$ .
- We focus on the **dynamics of the growth rates of  $A$  and  $K$** .
- Substituting the production function, (3.1), into the expression for capital accumulation, (3.3), yields

$$\dot{K}(t) = s(1 - a_k)^\alpha (1 - a_L)^{1-\alpha} K(t)^\alpha A(t)^{1-\alpha} L(t)^{1-\alpha} \quad (3.13)$$

Dividing both sides by  $K(t)$  and defining  $c_K = s(1 - a_k)^\alpha (1 - a_L)^{1-\alpha}$  gives us

$$\begin{aligned} g_K(t) &\equiv \frac{\dot{K}(t)}{K(t)} \\ &= c_K \left[ \frac{A(t)L(t)}{K(t)} \right]^{1-\alpha} \end{aligned} \quad (3.14)$$

Taking logs of both sides and differentiating with respect to time yields

$$\frac{g_K(t)}{g_K(t)} = (1 - \alpha)[g_A(t) + n - g_K(t)] \quad (3.15)$$

- From (3.13),  $g_K$  is always positive.
  - Thus  $g_K$  is **rising** if  $g_A + n - g_K$  is **positive**,
  - **falling** if this expression is **negative**, and
  - **constant** if it is **zero**.

This information is summarized in Figure 3.5. In  $(g_A, g_K)$  space,

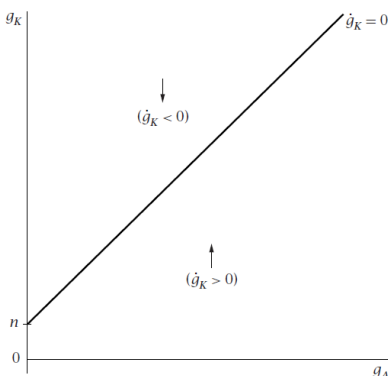


FIGURE 3.5 The dynamics of the growth rate of capital in the general version of the model

- The locus of points where  $g_K$  is constant has an **intercept** of  $n$  and a **slope** of  $1$ .  
[Because  $g_K = n + 1 \cdot g_A$ ]
- **Above** the locus,  $g_K$  is **falling**
- **below** the locus, it is **rising**

Similarly, dividing both sides of equation (3.2),  $\dot{A} = B(a_K K)^\beta (a_L L)^\gamma A^\theta$  by  $A$  yields an expression for the growth rate of  $A$ :

$$g_A(t) = c_A K(t)^\beta L(t)^\gamma A(t)^{\theta-1} \quad (3.16)$$

where  $c_A \equiv B a_K^\beta a_L^\gamma$

Taking logs and differentiating with respect to time gives

$$\frac{g_A(t)}{g_A(t)} = \beta g_K(t) + \gamma n + (\theta - 1)g_A(t) \quad (3.17)$$

- Thus  $g_A$  is **rising** if  $\beta g_K + \gamma n + (\theta - 1)g_A$  is **positive**,
- **falling** if it is **negative**
- **constant** if it is **zero**

This is shown in Figure 3.6.

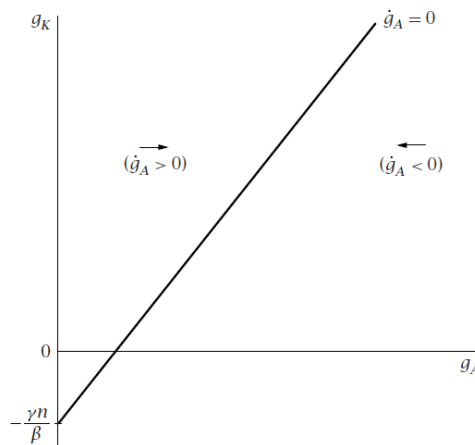


FIGURE 3.6 The dynamics of the growth rate of knowledge in the general version of the model

- The set of points where  $g_A$  is **constant** has an **intercept** of  $-\frac{\gamma n}{\beta}$  and a **slope** of  $\frac{1-\theta}{\beta}$   
[See explanation]
- **Above** this locus,  $g_A$  is **rising**
- **Below** the locus, it is **falling**

**Explanation:**

When  $g_A$  is constant:

$$\beta g_K + \gamma n + (\theta - 1)g_A = 0$$

$$\beta g_K = -\gamma n + (1 - \theta)g_A$$

$$g_K = \frac{-\gamma n + (1 - \theta)g_A}{\beta}$$

$$g_K = -\frac{\gamma n}{\beta} + \frac{1 - \theta}{\beta} g_A$$

Thus, the **intercept** is  $-\frac{\gamma n}{\beta}$

The **slope** is  $\frac{1 - \theta}{\beta}$

- ❖ The production function **for output** (equation [3.1]) exhibits **constant returns** to scale in the two produced factors of production, capital and knowledge.
  - Thus whether there are **on net increasing**, **decreasing**, or **constant** returns to scale to the produced factors depends on
    - their **returns to scale** in the **production function for knowledge**, equation (3.2).
    - As that equation shows, the **degree of returns to scale** to  $K$  and  $A$  in knowledge production is  $\beta + \theta$

- Thus the key determinant of the economy's behavior is **how  $\beta + \theta$  compares with 1**
- We will limit our attention to the cases of
  - i)  $\beta + \theta < 1$  and of
  - ii)  $\beta + \theta = 1$  with  $n = 0$

The remaining cases ( $\beta + \theta > 1$  and  $\beta + \theta = 1$  with  $n > 0$ ) have implications **similar** to those of  $\theta > 1$  in the **simple model**.

### Case 1: $\beta + \theta < 1$

- If  $\beta + \theta < 1$ , then  $\frac{1-\theta}{\beta} > 1$ .
- Thus the locus of points where  $\dot{g}_A = 0$  is **steeper** than the locus where  $\dot{g}_K = 0$

#### Explanation:

$$\begin{aligned} \beta + \theta &< 1 \\ \beta &< 1 - \theta \\ 1 - \theta &> \beta \\ \frac{1-\theta}{\beta} &> 1 \end{aligned}$$

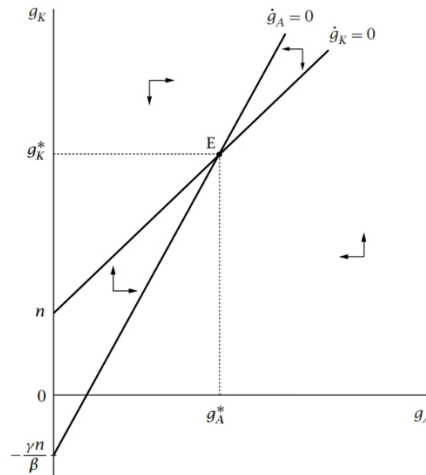


FIGURE 3.7 The dynamics of the growth rates of capital and knowledge when  $\beta + \theta < 1$

- Figure 3.7 shows that regardless of where  $g_A$  and  $g_K$  begin,
  - they **converge to Point E** in the diagram.
  - Both  $\dot{g}_A$  and  $\dot{g}_K$  are **zero** at this point.
- Thus the values of  $g_A$  and  $g_K$  at Point E, which we denote  $g_A^*$  and  $g_K^*$  must satisfy

$$g_A^* + n - g_K^* = 0 \quad (3.18)$$

and

$$\beta g_K^* + \gamma n + (\theta - 1) g_A^* = 0 \quad (3.19)$$

Rewriting (3.18) as  $g_K^* = g_A^* + n$  and substituting into (3.19) yields

$$\beta g_A^* + (\beta + \gamma)n + (\theta - 1)g_A^* = 0 \quad (3.20)$$

or

$$g_A^* = \frac{\beta + \gamma}{1 - (\theta + \beta)} n \quad (3.21)$$

**Explanation:**

$$\beta(g_A^* + n) + \gamma n + (\theta - 1)g_A^* = 0$$

$$\beta g_A^* + \beta n + \gamma n + (\theta - 1)g_A^* = 0$$

$$\beta g_A^* + (\beta + \gamma)n + (\theta - 1)g_A^* = 0$$

$$\beta g_A^* + (\theta - 1)g_A^* = -(\beta + \gamma)n$$

$$g_A^* [\beta + (\theta - 1)] = -(\beta + \gamma)n$$

$$g_A^* = -\frac{(\beta + \gamma)n}{[\beta + (\theta - 1)]}$$

$$g_A^* = -\frac{(\beta + \gamma)n}{-(\beta - \theta + 1)}$$

$$g_A^* = \frac{\beta + \gamma}{1 - (\theta + \beta)} n$$

- From above,  $g_K^*$  is simply  $g_A^* + n$
- Equation (3.1) then implies that
  - when  $A$  and  $K$  are **growing at these rates**, **output** is **growing** at rate  $g_K^*$
  - **Output per worker** is therefore **growing** at rate  $g_A^*$
- This case is **similar** to the case
  - when  $\theta$  is **less than 1** in the version of the **model without capital**.

Here, as in that case,

- **the long-run growth rate** of the economy is **endogenous**, and
- again **long-run growth** is an **increasing** function of **population growth**
  - and is **zero** if population growth is **zero**
- The fractions of the labor force and the capital stock engaged in R&D,
  - $a_L$  and  $a_K$ , **do not affect long-run growth**;
  - **nor does** the saving rate,  $s$ .
- The reason that these parameters do not affect long-run growth is essentially the **same** as the **reason** that  $a_L$  does not affect long-run growth in the simple version of the model.

❖ Models like **this one** and like the **model without capital** in the case of  $\theta < 1$  are often referred to as **semi-endogenous growth models**.

- On the one hand, **long-run growth** arises **endogenously** in the model.

- On the other, it **depends only on population growth** and **parameters of the knowledge production function**,
- and is **unaffected** by any **other parameters** of the model.
- Thus, as the name implies, growth seems only **somewhat endogenous**.

### Case 2: $\beta + \theta = 1$ and $n = 0$

- We have seen that the **locus of points** where  $\dot{g}_K = 0$  is given by  $g_K = g_A + n$
- and that the **locus of points** where  $\dot{g}_A = 0$  is given by  $g_K = -\frac{\gamma n}{\beta} + \frac{1-\theta}{\beta} g_A$
- When  $\beta + \theta = 1$  and  $n=0$ , both expressions simplify to  $g_K = g_A$

#### Explanation:

#### When $\beta + \theta = 1$ and $n=0$

$$\begin{aligned} \text{If, } \beta + \theta &= 1 \\ \beta &= 1 - \theta \\ \frac{1-\theta}{\beta} &= 1 \dots\dots\dots(i) \end{aligned}$$

- **the locus of  $\dot{g}_K = 0$**

$$\begin{aligned} g_K &= g_A + n \text{ can be written as:} \\ g_K &= g_A \quad (\text{since, } n=0) \end{aligned}$$

- **the locus of  $\dot{g}_A = 0$**

$$\begin{aligned} g_K &= -\frac{\gamma n}{\beta} + \frac{1-\theta}{\beta} g_A \text{ can be written as:} \\ g_K &= \frac{1-\theta}{\beta} g_A \quad (\text{since, } n=0) \\ g_K &= g_A \quad [\text{substituting the value from (i)}] \end{aligned}$$

- That is, in this case the **two loci lie directly on top of each other**:
  - both are given by the 45-degree line.
  - Figure 3.8 shows the dynamics of the economy in this case.

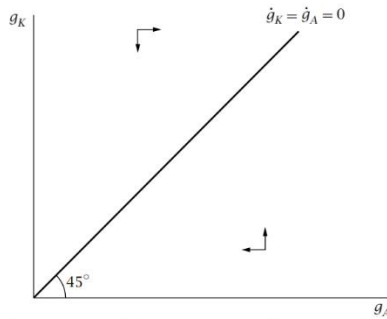
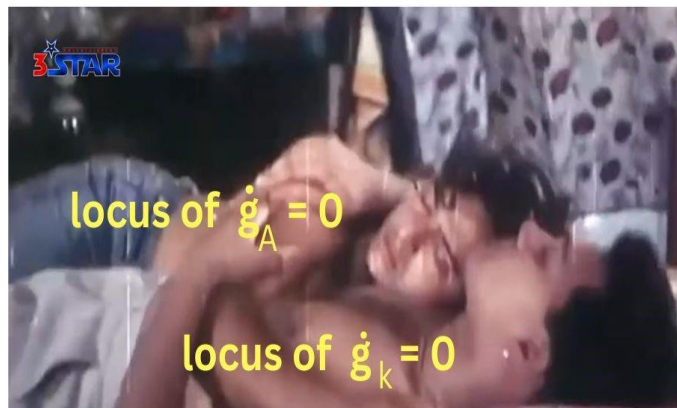


FIGURE 3.8 The dynamics of the growth rates of capital and knowledge when  $\beta + \theta = 1$  and  $n = 0$

- As the figure shows, **regardless of where the economy begins**,
  - the dynamics of  $g_A$  and  $g_K$  carry them **to the 45-degree line**.
  - Once that happens,  $g_A$  and  $g_K$  are **constant**
  - and the economy is on a **BGP**.

### R&D model with capital when $\beta + \theta = 1$ and $n = 0$



- As in the case of  $\theta = 1$  and  $n = 0$  in the model **without capital**
  - the phase diagram **does not tell us** **what BGP the economy converges to**
- However one can show that
  - the economy has a **unique BGP** for a given set of parameter values
  - Increases** in the **saving rate** and in the size of the population **increase** this **long-run growth rate**
  - And because **changes** in  $a_L$  and  $a_K$  involve **shifts** of resources between goods production and R&D, they have **ambiguous effects** on **long-run growth**



Because **long-run growth** depends on a wide range of parameters, models like **this one**, as well as the model of the **previous section** when  $\theta \geq 1$  and the model of **this section** when  $\beta + \theta > 1$  or  $\beta + \theta = 1$  and  $n > 0$ , are known as *fully endogenous growth models*.

### 3.4 The Nature of Knowledge and the Determinants of the Allocation of Resources to R&D

**This part does not look much important for the exam. But you should give it a read. Life is unpredictable.**

#### Overview

- This section is devoted to the discussion of **what determines**  $a_L$  and  $a_K$
- So far we have simply described
  - the “ $A$ ” variable produced by R&D as knowledge.
- But knowledge comes in **many forms**
- Many of these **different types** of knowledge
  - play important roles in economic growth.
- There is no reason to expect
  - the determinants of the accumulation of these **different types of knowledge** to be the same
- we should expect to find **various factors** underlying the accumulation of knowledge.
- At the same time, all types of knowledge share one essential feature:
  - they are **nonrival**
  - That is, the use of an item of knowledge, in one application makes its use by someone else no more difficult
- An immediate implication of this fundamental property of knowledge is that
  - the production and allocation of knowledge **cannot be** completely governed by competitive market forces.
- Although all knowledge is **nonrival**
  - it is **heterogeneous** along a second dimension: **excludability**.
  - A good is excludable if it is possible to prevent others from using it.
- In the case of knowledge, **excludability depends** both on the
  - nature of the knowledge itself
  - and on economic institutions governing property rights.
  - Patent laws, for example, give inventors rights over the use of their designs and discoveries.
- In some cases, excludability is
  - more dependent on the nature of the knowledge
  - and less dependent on the legal system.

- The recipe for Coca-Cola is sufficiently complex that it can be kept secret without copyright or patent protection.
- The degree of excludability is likely to have a strong influence on
  - how the development and allocation of knowledge depart from perfect competition.
- If a type of knowledge is **entirely nonexcludable**,
  - there can be **no private gain** in its development;
  - thus R&D in these areas must come from **elsewhere**.
- But when knowledge is **excludable**
  - the producers of new knowledge can license the right to use the knowledge at positive prices,
  - and hence hope to **earn positive returns** on their R&D efforts.
- ❖ With these broad remarks, we can now turn to a discussion of some of the **major forces** governing the allocation of resources to the **development of knowledge**
- ❖ **Four forces** have received the most attention:
  - Support for basic scientific research
  - Private incentives for R&D and innovation,
  - Alternative opportunities for talented individuals, and
  - Learning-by-doing.

### Support for Basic Scientific Research

- **Basic scientific knowledge** has traditionally been **made available relatively freely**
- This research is **not motivated by** the desire to **earn private returns** in the market.
- Instead it is supported by governments, charities, and wealthy individuals
- Since it is useful in production and is given away at zero cost
  - it has a **positive externality**

### Private Incentives for R&D and Innovation

- Many innovations are motivated almost entirely by the **desire for private gain**
- For R&D to result from economic incentives,
  - the knowledge that is created must be at least **somewhat excludable**.
  - Thus the developer of a new idea has some degree of market power
- Since economies like these are not perfectly competitive, their equilibria are not in general optimal.
- In particular, the decentralized equilibria may have inefficient divisions of resources between R&D and conventional goods production.
- There are in fact **three distinct externalities** from R&D:
  - the *consumer-surplus* effect

- the *business-stealing* effect, and
- the *R&D* effect
- **The consumer-surplus effect** is that the individuals or firms licensing ideas from innovators obtain some surplus, since innovators cannot engage in perfect price discrimination. Thus this is a **positive externality** from R&D.
- **The business-stealing effect** is that the introduction of a superior technology typically makes existing technologies less attractive, and therefore harms the owners of those technologies. This externality is **negative**.
- Finally, the **R&D effect** is that innovators are generally assumed not to control the use of their knowledge in the production of additional knowledge. Thus the development of new knowledge has a **positive externality** on others engaged in R&D.
- The **net effect** of these three externalities is **ambiguous**.
  - It is possible to construct examples where the business-stealing externality outweighs both the consumer-surplus and R&D externalities.
  - In this case the incentives to capture the profits being earned by other innovators cause too many resources to be devoted to R&D.
- However, it is generally believed, that
  - **the normal situation** is for the overall externality from R&D to be **positive**.
- There can be additional externalities as well.
  - For example, if innovators have only **incomplete control** over the use of their ideas in goods production there is an additional reason that the private return to R&D is below the social return.
  - On the other hand, the fact that the **first individual to create an invention** is awarded exclusive rights to the invention can create excessive incentives for some kinds of R&D;

### Alternative Opportunities for Talented Individuals

- Baumol (1990) and Murphy, Shleifer, and Vishny (1991) observe that
  - **major innovations** and **advances in knowledge** are often the result of the **work of extremely talented individuals**

### Ignore this list

Baumol argues that, in various places and times

- military conquest
- political and religious leadership
- tax collection
- criminal activity
- philosophical contemplation
- financial dealings and
- manipulation of the legal system

have been attractive to the most talented members of society.

- Baumol argues that activities of the most talented members of society
  - often have **negligible** (or even **negative**) **social returns**
  - these activities are often forms of **rent-seeking**—attempts to capture existing wealth rather than to create new wealth.
  - there has been a strong link between how societies direct the energies of their most able members and whether the societies flourish over the long term.

Murphy, Shleifer, and Vishny provide a general discussion of the

- forces that influence talented individuals' decisions
  - whether to pursue activities that are socially productive

They emphasize **three factors** in particular.

- The first is the **size of the relevant market**
  - the **larger** is the **market** from which a talented individual can reap returns
  - the **greater are the incentives** to enter a given activity
- The second factor is the **degree of diminishing returns**
  - Activities whose scale is limited by the entrepreneur's time
  - do not offer the same potential returns
  - as activities whose returns are limited only by the scale of the market
- The final factor is the **ability to keep the returns** from one's activities.
  - Thus, **clear property rights** tend to encourage entrepreneurship

### Learning-by-Doing

According to Arrow (1962)

- the accumulation of knowledge occurs in part not as a result of deliberate efforts
  - but as a **side effect of conventional economic activity**.
- This type of knowledge accumulation is known as **learning-by-doing**

When learning-by-doing is the source of technological progress

- the rate of knowledge accumulation
  - depends on **how much new knowledge is generated** by conventional economic activity.

Analyzing learning-by-doing therefore requires some changes to our model.

- All inputs are now engaged in **goods production**
- thus the production function becomes

$$Y(t) = K(t)^\alpha [A(t)L(t)]^{1-\alpha} \quad (3.22)$$

With this formulation

- the **increase in knowledge** is a function of the **increase in capital**
- the stock of **knowledge** is a function of the stock of **capital**
- Thus there is only one state variable
- Making our usual choice of a power function, we have

$$A(t) = K(t)^\alpha, \quad B > 0, \quad \varphi > 0. \quad (3.23)$$

To analyze the economy, we begin by substituting (3.23) into (2.22)

This yields

$$Y(t) = K(t)^\alpha B^{1-\alpha} K(t)^{\varphi(1-\alpha)} L(t)^{1-\alpha} \quad (3.24)$$

Since  $\dot{K}(t) = s Y(t)$ , the dynamics of  $K$  are given by

$$\dot{K}(t) = s B^{1-\alpha} K(t)^\alpha K(t)^{\varphi(1-\alpha)} L(t)^{1-\alpha} \quad (3.25)$$

In our model of knowledge accumulation without capital in Section 3.2,

- the dynamics of  $A$  are given  $\dot{A}(t) = B[a_L L(t)]^\nu A(t)^\theta$  (equation [3.6])

Comparing equation (3.25) of the learning-by-doing model with this equation shows that

- the **structures** of the two models are **similar**.
- the **dynamics** of the two models are essentially the **same**.

Thus we can use the results of our analysis of the earlier model to analyze this one.

- In the model of knowledge accumulation without capital
    - the key determinant of the economy's dynamics is **how  $\theta$  compares with 1**.
  - Here, by analogy, it is **how  $\alpha + \varphi(1 - \alpha)$  compares with 1**
    - which is equivalent to **how  $\varphi$  compares with 1**
- If  $\varphi < 1$ , the **long-run growth rate** of the economy is a **function of the rate of population growth,  $n$** .
  - If  $\varphi > 1$ , there is **explosive growth**
  - if  $\varphi = 1$ , there is **explosive growth** if  $n > 0$  and **steady growth** if  $n = 0$

- Once again, a case that has received particular attention is  $\varphi = 1$  and  $n = 0$
- In this case, the production function (equation [3.24]) becomes

$$Y(t) = bK(t), \quad b \equiv B^{1-\alpha} L^{1-\alpha} \quad (3.26)$$

Capital accumulation is therefore given by

$$\dot{K}(t) = sbK(t) \quad (3.27)$$

The dynamics of this economy are straightforward

- Equation (3.27) immediately implies that  **$K$  grows steadily** at rate  $sb$
  - And since **output** is proportional to  $K$ , it **also grows at this rate**.
- ❖ Thus we have another example of a model in which **long-run growth is endogenous** and **depends on the saving rate**
- In this model, the saving rate affects long-run growth
    - because the **contribution of capital** is **larger than its conventional contribution**
  - **increased capital raises output** not only through its direct role in production
    - but also by **indirectly contributing** to the **development of new ideas**
    - and thereby making all other capital more productive

Because the production function in these models is often written using the symbol “ $A$ ” rather than the “ $b$ ” used in (3.26), these models are often referred to as “ **$Y = AK$ ” models**.